Single-Cell Reanalysis of HSV Infection with R/Seurat

Data source

Paper and author-deposited data

- **Original paper — Lebratti et al. (2021)** — Experimental methods, Figure 3 and the stated PMN markers.
- **GEO: GSE161336** — Seven processed single-cell UMI matrices, with barcodes and gene features: mock and HSV-1/HSV-2 at days 1, 3 and 5.
- **ENA / SRA: PRJNA677918** — Author-deposited sequencing files and run metadata from the same study. We used these for read-count summaries and viral-alignment checks.
- **Figure 3 source data 1** — Supplementary neutrophil differential-expression results and Figure 3E IFN-beta ELISA measurements. These were downloaded for reference. Figure 3E is outside this report's current scope.

Additional sources used in our analysis

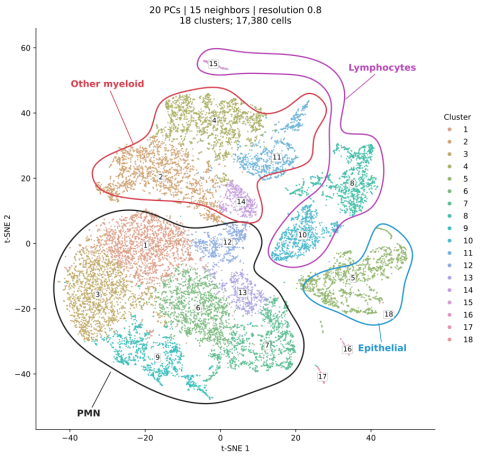
- **MSigDB: Hallmark IFN-alpha response** — The paper names this gene set. We downloaded a current mouse version for Figure 3D scoring (89/94 genes matched). Its equivalence to the paper's historical version is unverified.
- **PanglaoDB markers** — Additional candidate markers for other myeloid, epithelial and lymphocyte clusters. These annotations remain provisional.
- **NCBI viral references: HSV-1 NC_001806.2 and HSV-2 JN561323.2** — Reference records we identified from the deposited BAM headers. These accessions are not explicitly named in the paper's methods.

High-level steps

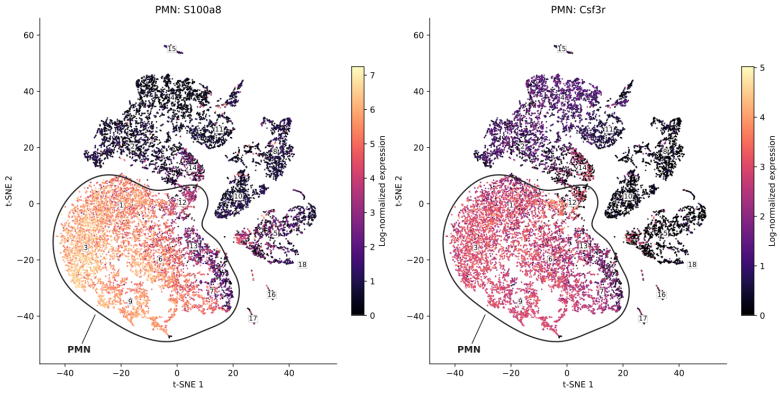
1. **Preprocess**
   - Load and combine the seven samples.
   - Filter high-mitochondrial cells and low-gene clusters, then review sample QC.
2. **Prepare**
   - Normalize counts and select highly variable genes.
   - Regress total UMI, scale and run PCA.
3. **Clustering**
   - Build the neighbor graph, cluster cells and plot t-SNE.
   - Annotate cell types with markers and compare conditions.
   - Plot IFN-alpha response scores.

Step-by-step reproduction

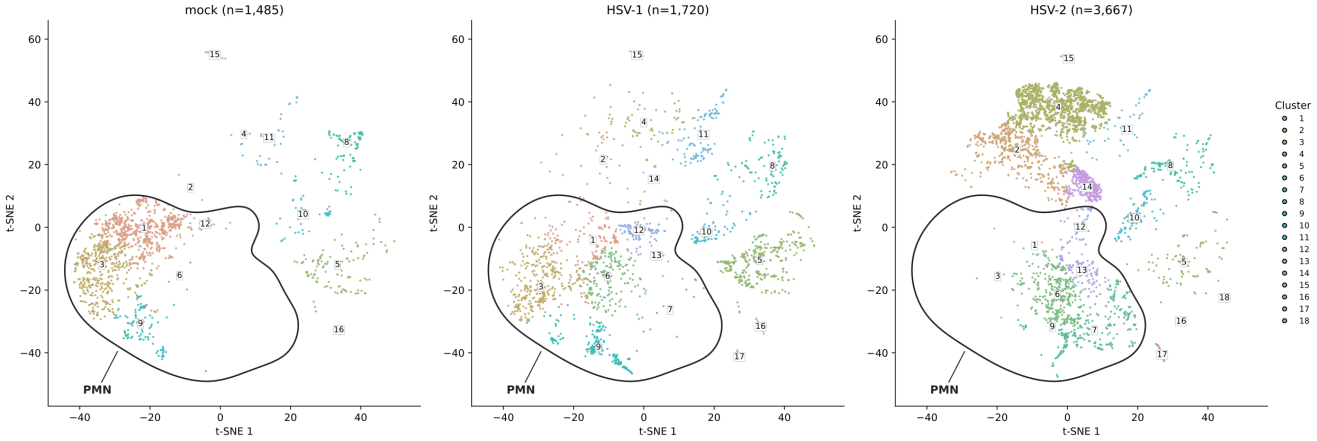
A



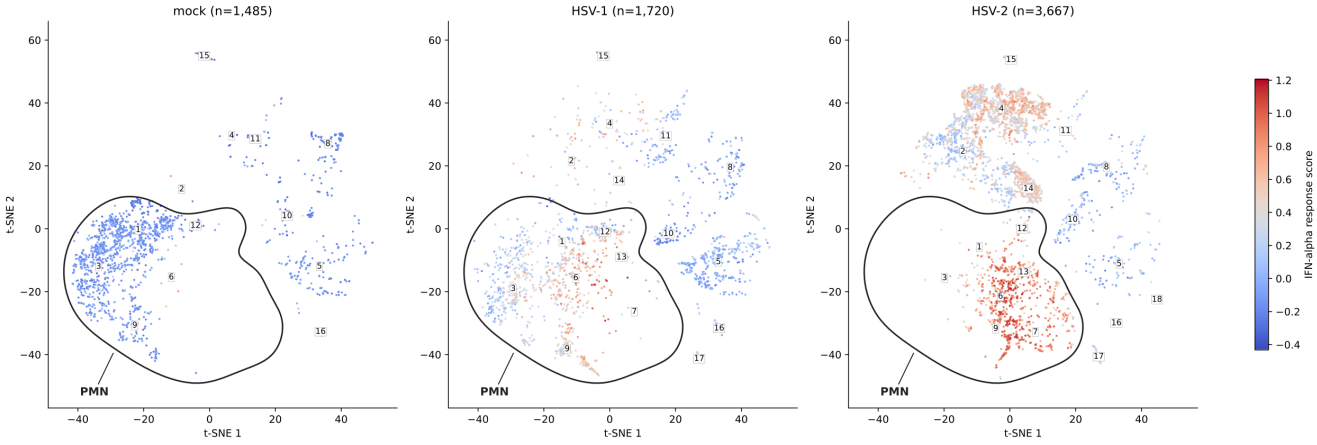
B



C



D



1. Preprocess

1.1 Load the seven samples

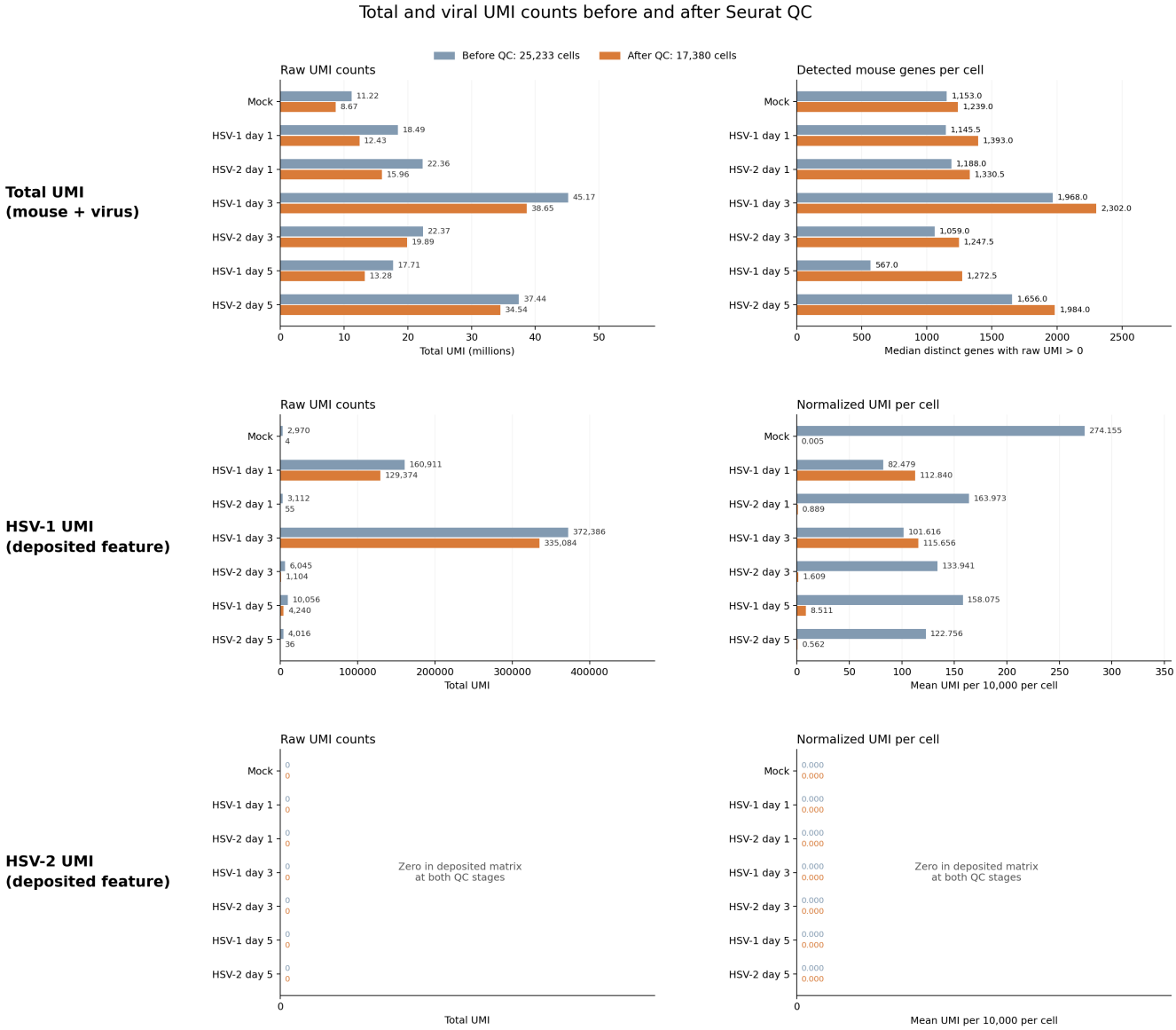
Combine mock and HSV-1/HSV-2 samples at days 1, 3 and 5: **25,233 cells** and **31,055 features**.

1.2 Filter low-quality cells

1. Remove cells with >5% mitochondrial UMI: **20,795 cells remain**.
2. Normalize, select 3,000 HVGs, scale and run initial clustering with **30 PCs, 15 neighbors and resolution 0.5**.
3. Remove clusters with median detected genes <500: **17,380 cells remain**.

Per-sample QC summary

Sample	Condition	Total reads (ENA)	Before QC			After QC (Seurat)		
			Cells	UMI	Median mouse genes/cell	Cells	UMI	Median mouse genes/cell
GSM4905024	mock	349,393,689	2,119	11,217,519	1,153.0	1,485	8,674,113	1,239.0
GSM4905025	HSV-1 day 1	329,855,668	2,764	18,485,658	1,145.5	1,491	12,427,504	1,393.0
GSM4905026	HSV-2 day 1	364,128,507	3,300	22,356,332	1,188.0	2,196	15,955,237	1,330.5
GSM4905027	HSV-1 day 3	372,867,577	4,715	45,165,593	1,968.0	3,549	38,653,781	2,302.0
GSM4905028	HSV-2 day 3	414,947,993	4,318	22,374,708	1,059.0	3,272	19,886,678	1,247.5
GSM4905029	HSV-1 day 5	371,353,516	3,488	17,709,707	567.0	1,720	13,279,914	1,272.5
GSM4905030	HSV-2 day 5	363,058,503	4,529	37,440,142	1,656.0	3,667	34,540,391	1,984.0
Total		**2,565,605,453**	**25,233**	**174,749,659**	—	**17,380**	**143,417,618**	—



Raw and normalized UMI, and detected mouse genes per cell, before and after QC.

2. Prepare

Start with the **17,380 retained cells**, keeping their original counts across all features. Recompute HVGs after QC because removing cells changes the expression variability across cells. paper methods, p.18.

2.1 Normalize counts

For every feature in every cell, Seurat's `NormalizeData(scale.factor = 10000)` performs:

1. Divide the feature's UMI count by that cell's total UMI across **all features**, not just HVGs.
2. Multiply by **10,000**, the reference total specified by the paper.
3. Apply the natural-log transformation **`log1p(x) = ln(1 + x)`**.

```
normalized expression = ln(1 + 10,000 × feature UMI / total cell UMI)
```

2.2 Select highly variable genes

Use `FindVariableFeatures(selection.method = 'mean.var.plot')` across the retained cells from all seven samples:

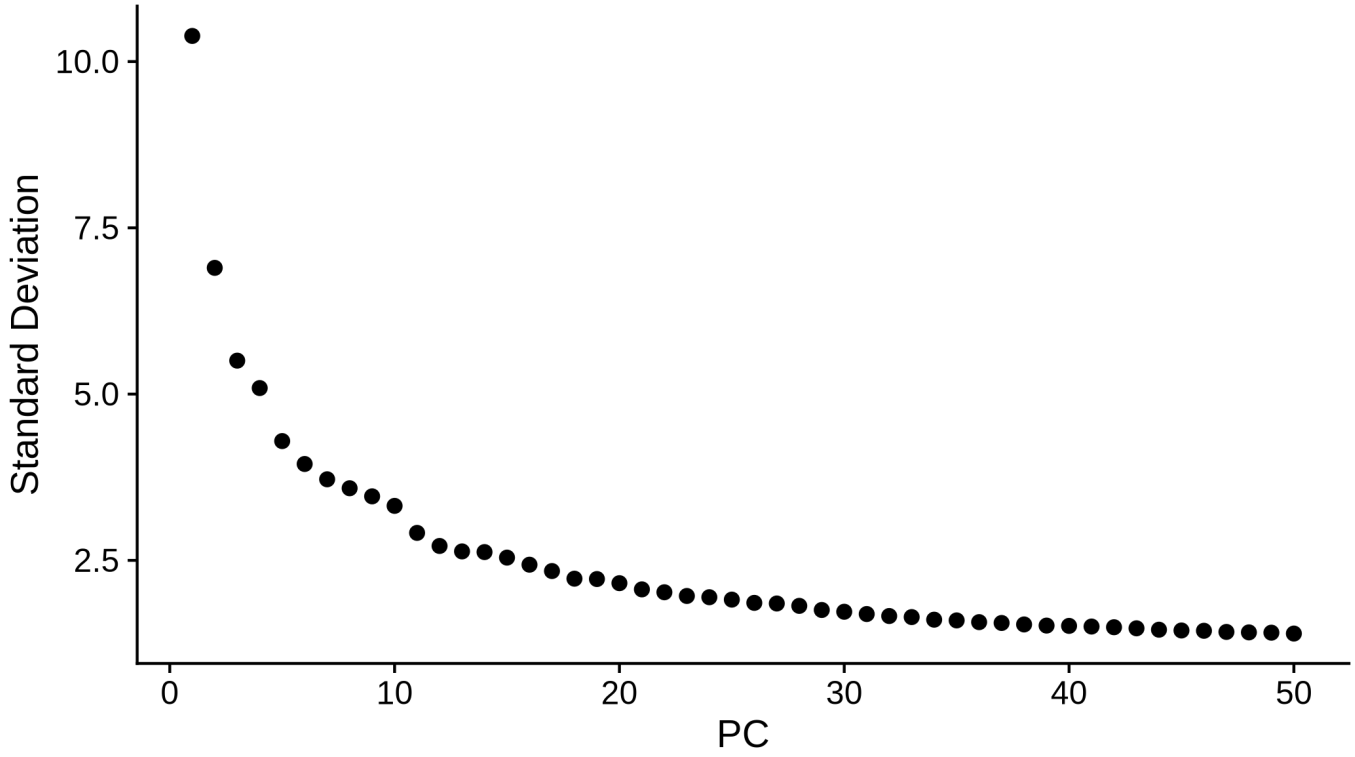
1. Calculate each feature's average expression and dispersion.
2. Group features into **20 expression bins** and standardize dispersion within each bin, comparing genes with similar average expression.
3. Rank by `mvp.dispersion.scaled` and explicitly select the **top 3,000 HVGs** for regression and PCA.

2.3 Regress total UMI and scale

For each HVG, model its **raw UMI counts** against each cell's total UMI (`nCount_RNA`), using `ScaleData(model.use = 'negbinom', use.umi = TRUE)`. This aims to reduce variation associated with total UMI. Infection condition is not included as a regression covariate.

2.4 Run PCA

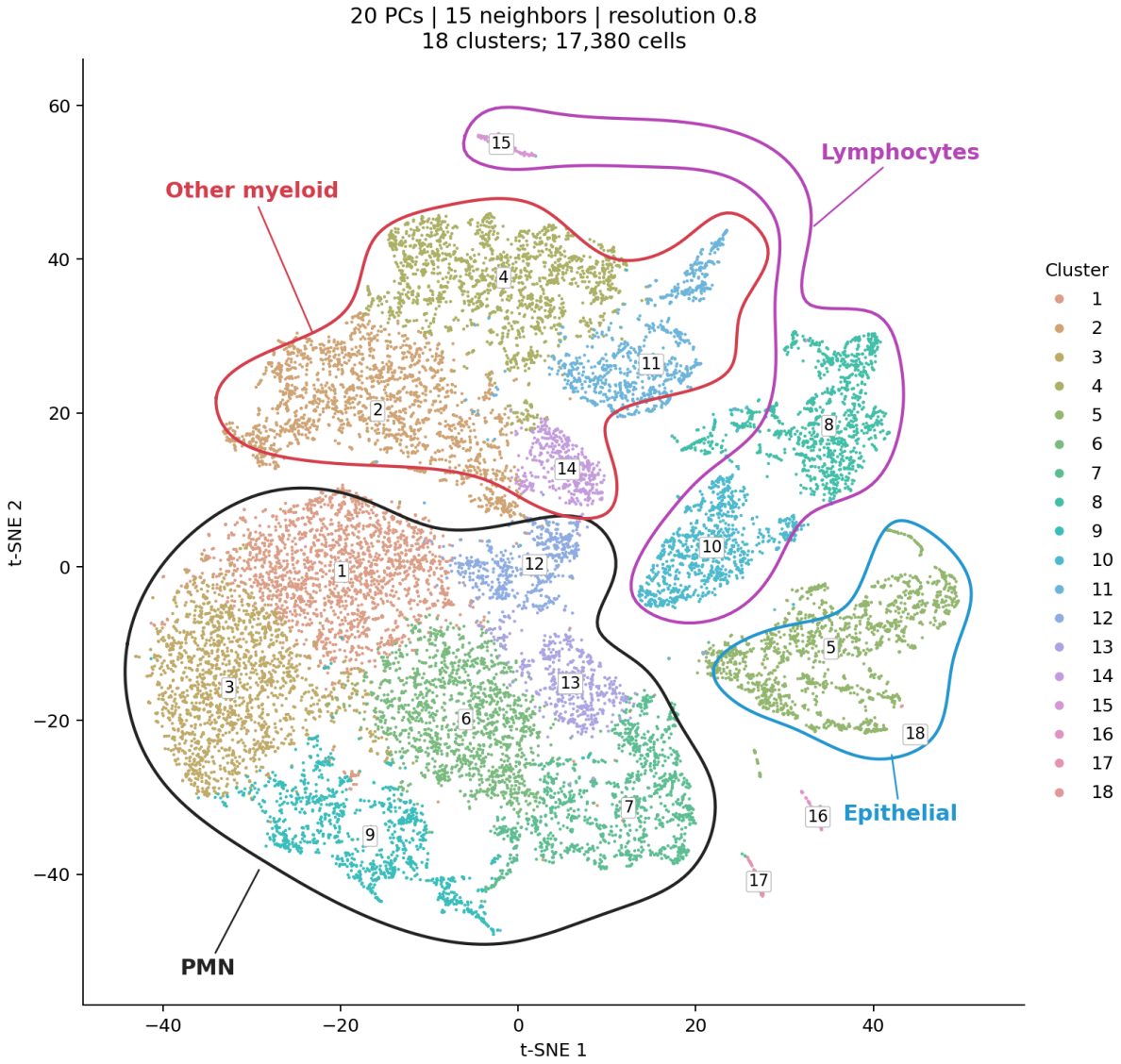
Use `RunPCA` on the adjusted **3,000 HVGs**, computing **50 principal components** with seed **42**. Each PC summarizes a weighted combination of genes. Each cell receives a score on each PC. Use the **first 20 PCs** for the final neighbor graph and t-SNE. Computing 50 allows inspection with an elbow plot. The choice of 20 PCs is specific to this analysis. The paper does not report an exact value.



3. Clustering

3.1 Cluster and visualize cells — Figure 3A

Build the shared-nearest-neighbor (SNN) graph with **20 PCs and 15 neighbors**, then run Leiden at **resolution 0.8**, producing **18 clusters**. Use the selected PCs for t-SNE.



Candidate type	R cluster IDs
PMN	1, 3, 6, 7, 9, 12, 13
Other myeloid	2, 4, 11, 14
Epithelial	5, 18
Lymphocytes	8, 10, 15

Clusters **16 and 17** remain unassigned. All annotations are provisional. Outlines are visual guides rather than exact boundaries. The connected lymphocyte outline does not imply a trajectory. Extra marker panels outline the relevant type. Figures 3C/D outline PMN only.

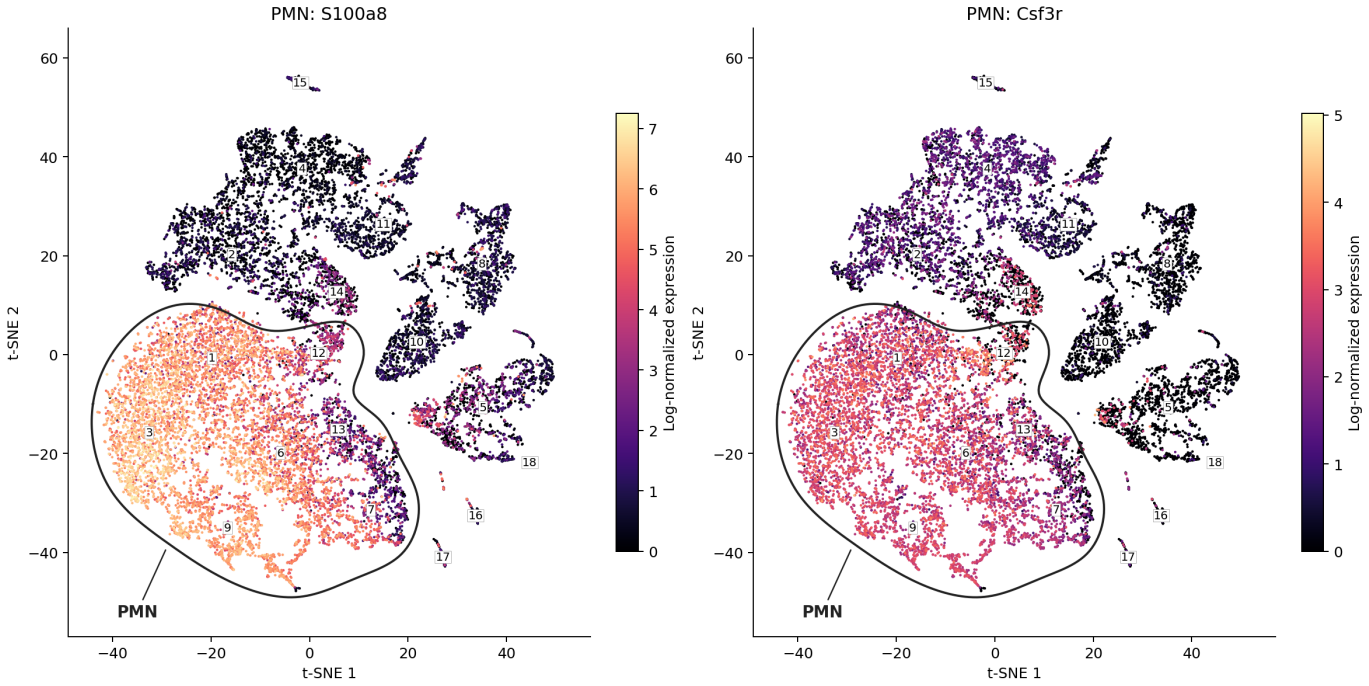
3.2 Annotate cell types — Figure 3B

Markers are used to interpret clusters after clustering. The paper specifies **S100a8/Csf3r** for neutrophils. The additional candidate markers support provisional annotations of other cell types.

Selected markers

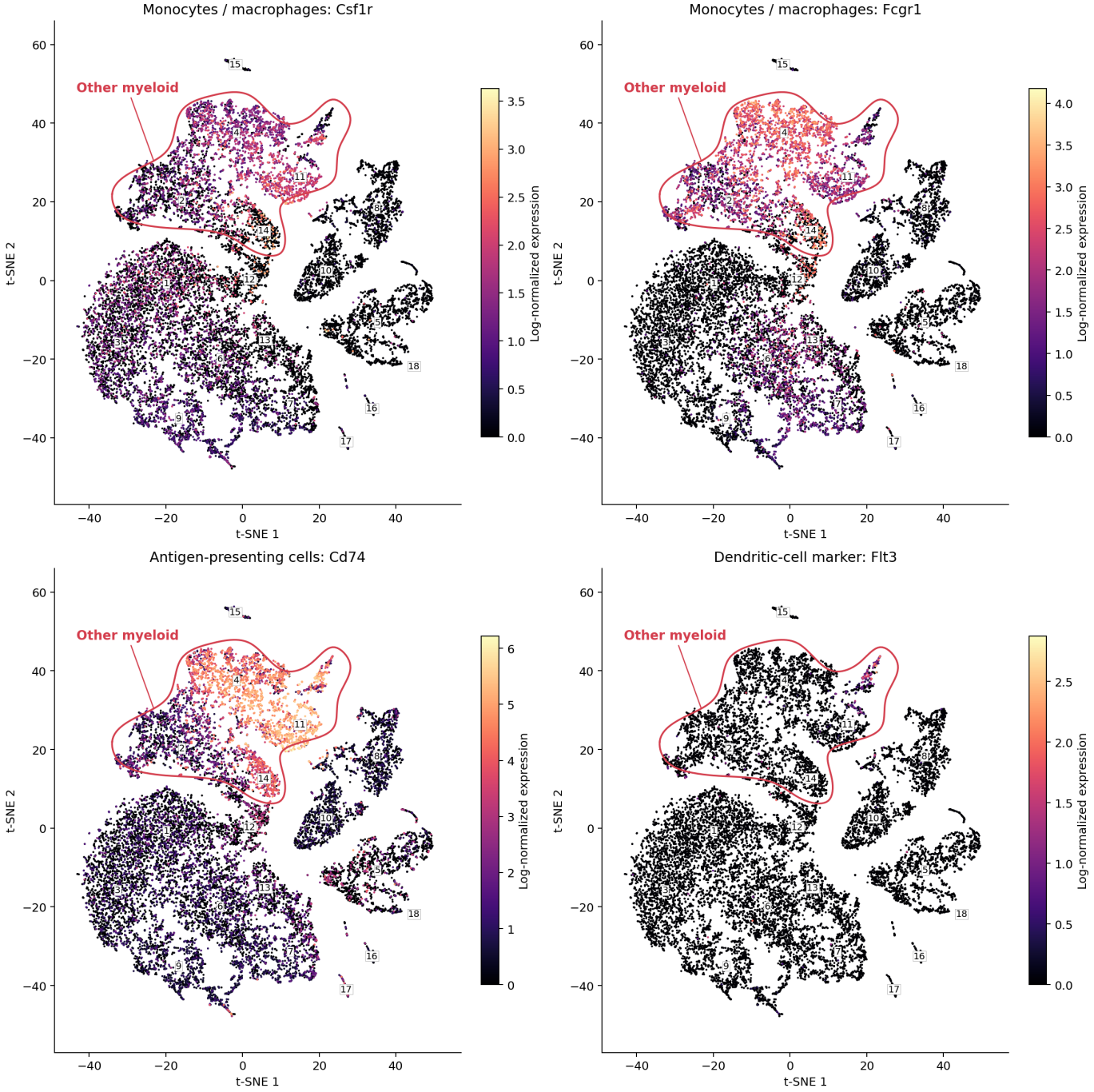
Group	Selected markers
PMN	S100a8, Csf3r
Other myeloid	Csf1r, Fcgr1 (monocytes/macrophages), Cd74, Flt3 (DC assessment)
Epithelial	Epcam, Krt5, Krt14, Krt18
Lymphocytes	Cd3d, Cd3e (T), Ncr1, Klrd1 (NK)

S100a8 and Csf3r log-normalized expression, with cluster IDs on both panels.

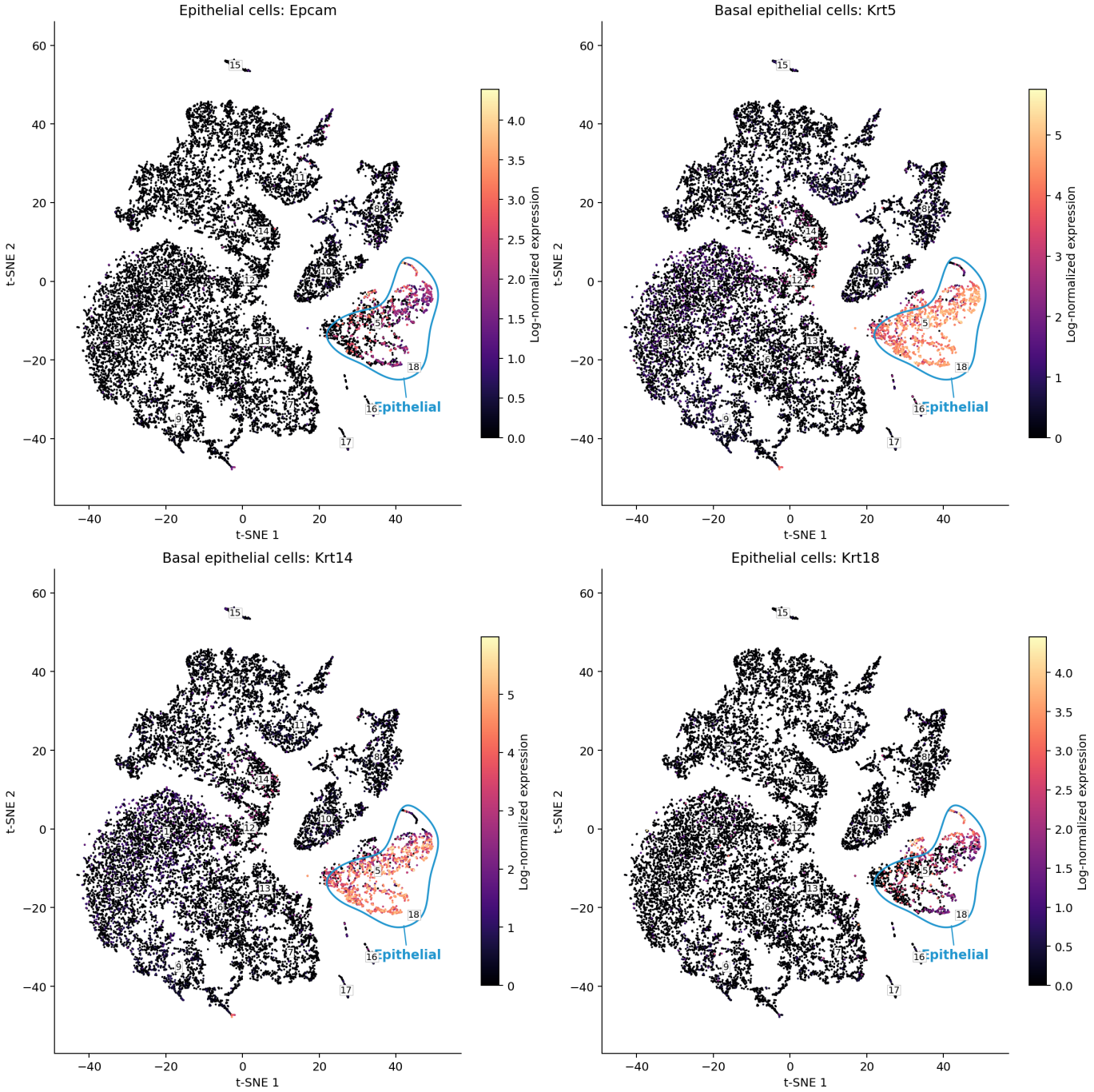


Extra: Markers for other cell types

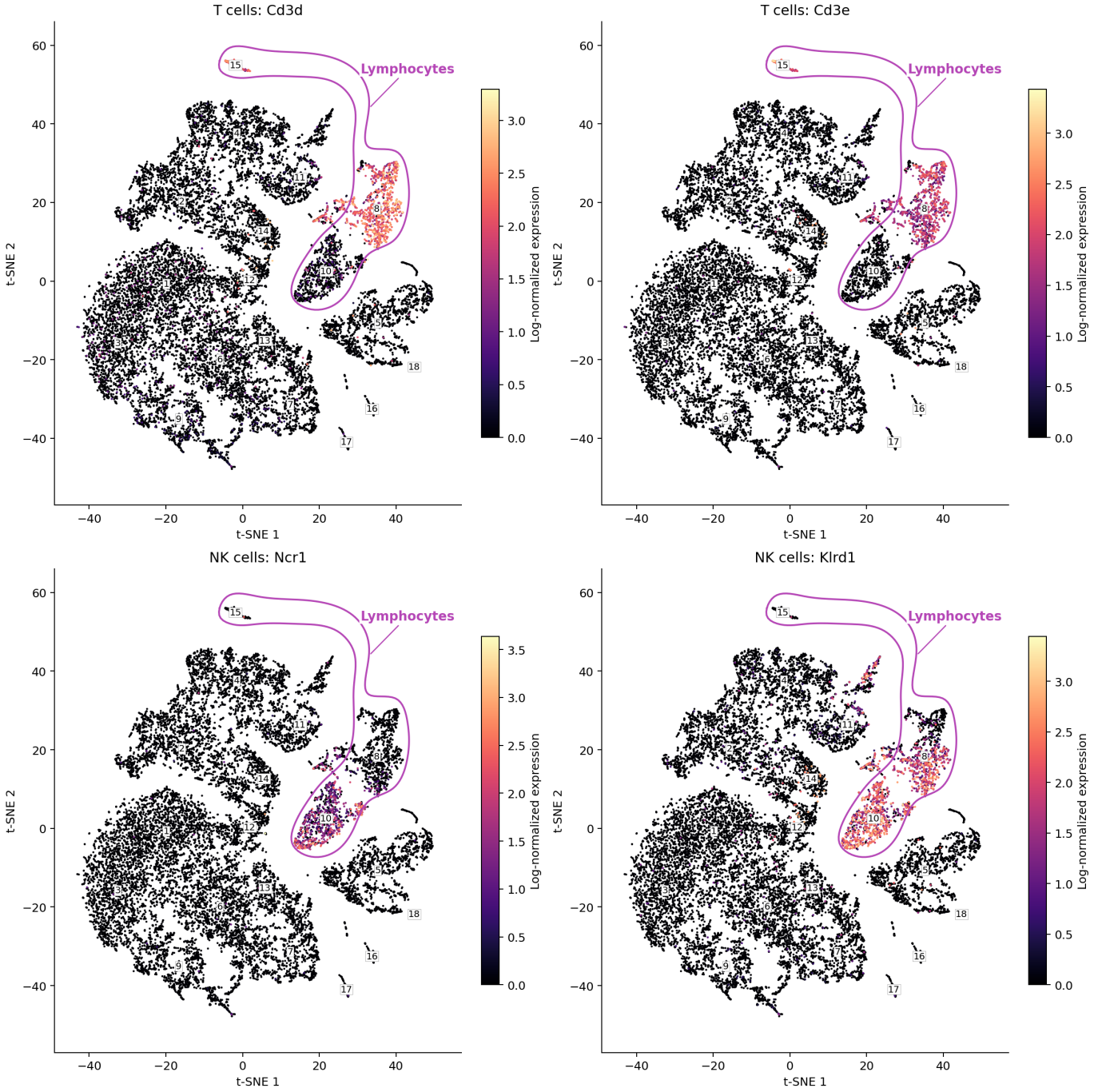
Other myeloid cells: Csf1r, Fcgr1, Cd74 and Flt3.



Epithelial cells: Epcam, Krt5, Krt14 and Krt18.

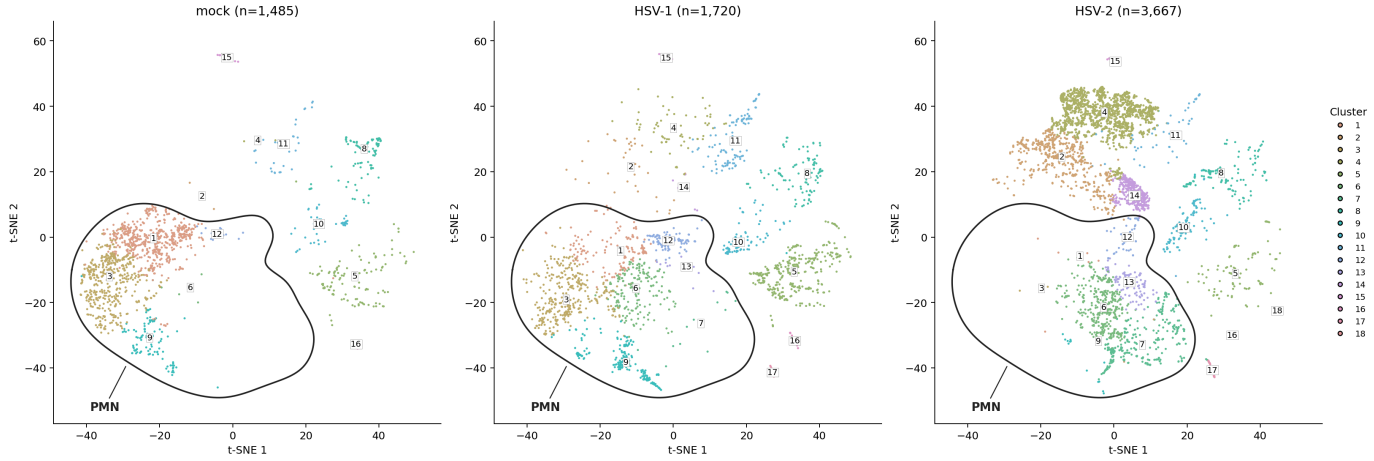


Lymphocytes: Cd3d/Cd3e for T cells, Ncr1/Klrd1 for NK cells.



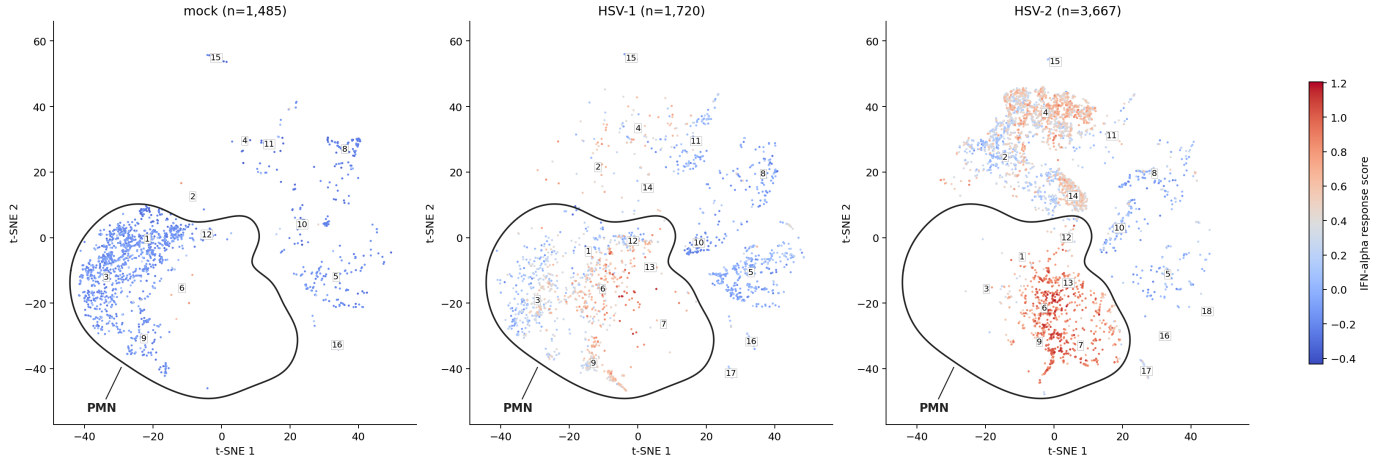
3.3 Compare conditions — Figure 3C

Display mock and day-5 HSV-1/HSV-2 cells on the shared seven-sample t-SNE, without reclustering.



3.4 Score the IFN-alpha response — Figure 3D

The same conditions colored by IFN-alpha response score.



Questions and answers

Sample-quality questions

The per-sample values are listed in **Step 1.2, Per-sample QC summary**.

Question	Answer
What is the total read count?	ENA reports 2,565,605,453 reads across the seven runs. These archive counts are distinct from UMI counts.
How many reads remain after QC?	Not calculated. We report UMI counts before and after our cell QC instead. Read counts after cell QC would require matching BAM reads to retained cell barcodes.
How many cells and UMIs remain after QC?	Cells: 25,233 → 17,380. Total UMI: 174,749,659 → 143,417,618.

Here, “before QC” means before our Seurat filtering. The deposited matrices have already undergone Cell Ranger cell calling.

Q1. What reference genomes did they use?

The paper specifies **S100a8/Csf3r** as neutrophil markers and examines **Cxcl10, Gbp2 and Il15**. The expression matrix contains only two viral features, **`HSV__HSV1` and `HSV__HSV2`**, without individual viral gene names.

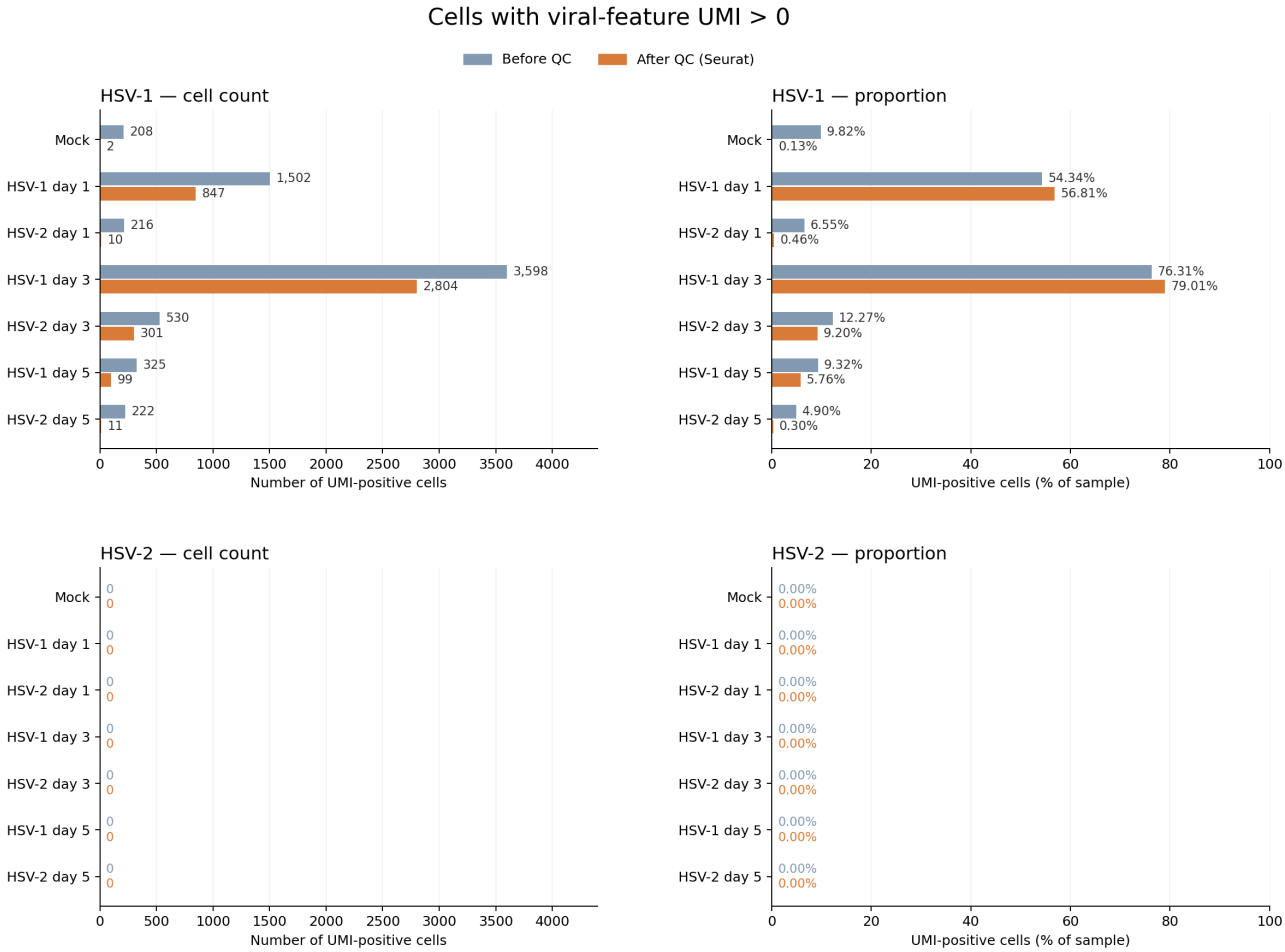
The table distinguishes these markers and features from the alignment references. GEO names the combined reference **`mm10_and_HSV`**, and the inspected BAM headers identify its viral sequences. GEO processing metadata.

Category	Reference or genes	Description
Mouse reference	mm10	Mouse component of the combined alignment reference
HSV-1 reference	NC_001806.2, strain 17	BAM reference `HSV__NC_001806.2`
HSV-2 reference	JN561323.2, strain HG52	BAM reference `HSV__JN561323.2`
Neutrophil markers	S100a8, Csf3r	Mouse genes used to identify PMN
Host-response genes	Cxcl10, Gbp2, Il15	Mouse genes examined in the paper
Viral matrix features	HSV__HSV1, HSV__HSV2	Two feature labels without individual viral gene names

Q2. How many cells in each sample were infected?

The paper does not provide a cell-level infection definition or counts. I temporarily defined **viral UMI >0** as positive and summarized the numbers and percentages of positive cells before and after QC.

Sample	Condition	Before QC		After QC (Seurat)	
		HSV-1 positive, n (%)	HSV-2 positive, n (%)	HSV-1 positive, n (%)	HSV-2 positive, n (%)
GSM4905024	mock	208 (9.82%)	0 (0.00%)	2 (0.13%)	0 (0.00%)
GSM4905025	HSV-1 day 1	1,502 (54.34%)	0 (0.00%)	847 (56.81%)	0 (0.00%)
GSM4905026	HSV-2 day 1	216 (6.55%)	0 (0.00%)	10 (0.46%)	0 (0.00%)
GSM4905027	HSV-1 day 3	3,598 (76.31%)	0 (0.00%)	2,804 (79.01%)	0 (0.00%)
GSM4905028	HSV-2 day 3	530 (12.27%)	0 (0.00%)	301 (9.20%)	0 (0.00%)
GSM4905029	HSV-1 day 5	325 (9.32%)	0 (0.00%)	99 (5.76%)	0 (0.00%)
GSM4905030	HSV-2 day 5	222 (4.90%)	0 (0.00%)	11 (0.30%)	0 (0.00%)



Percentages use all cells in the sample at the corresponding QC stage. Viral UMI positivity is a provisional infection proxy. The unexplained HSV-2 zeros and positive counts in mock are discussed in the gap analysis.

Q3. What percentage of each infected cell's reads came from the virus?

I calculated the **median viral UMI percentage among positive cells** for each sample. Each cell's percentage is `100 × viral UMI / total cell UMI`. This measures UMI percentage rather than read percentage. N/A means no cells were positive for that viral feature.

Sample	Condition	Before QC		After QC (Seurat)	
		HSV-1 median (%)	HSV-2 median (%)	HSV-1 median (%)	HSV-2 median (%)
GSM4905024	mock	23.5552	N/A	0.0340	N/A
GSM4905025	HSV-1 day 1	0.0328	N/A	0.0228	N/A
GSM4905026	HSV-2 day 1	19.5341	N/A	0.0956	N/A
GSM4905027	HSV-1 day 3	0.0257	N/A	0.0226	N/A
GSM4905028	HSV-2 day 3	0.1190	N/A	0.0426	N/A
GSM4905029	HSV-1 day 5	7.8818	N/A	0.0224	N/A
GSM4905030	HSV-2 day 5	18.4698	N/A	0.0123	N/A

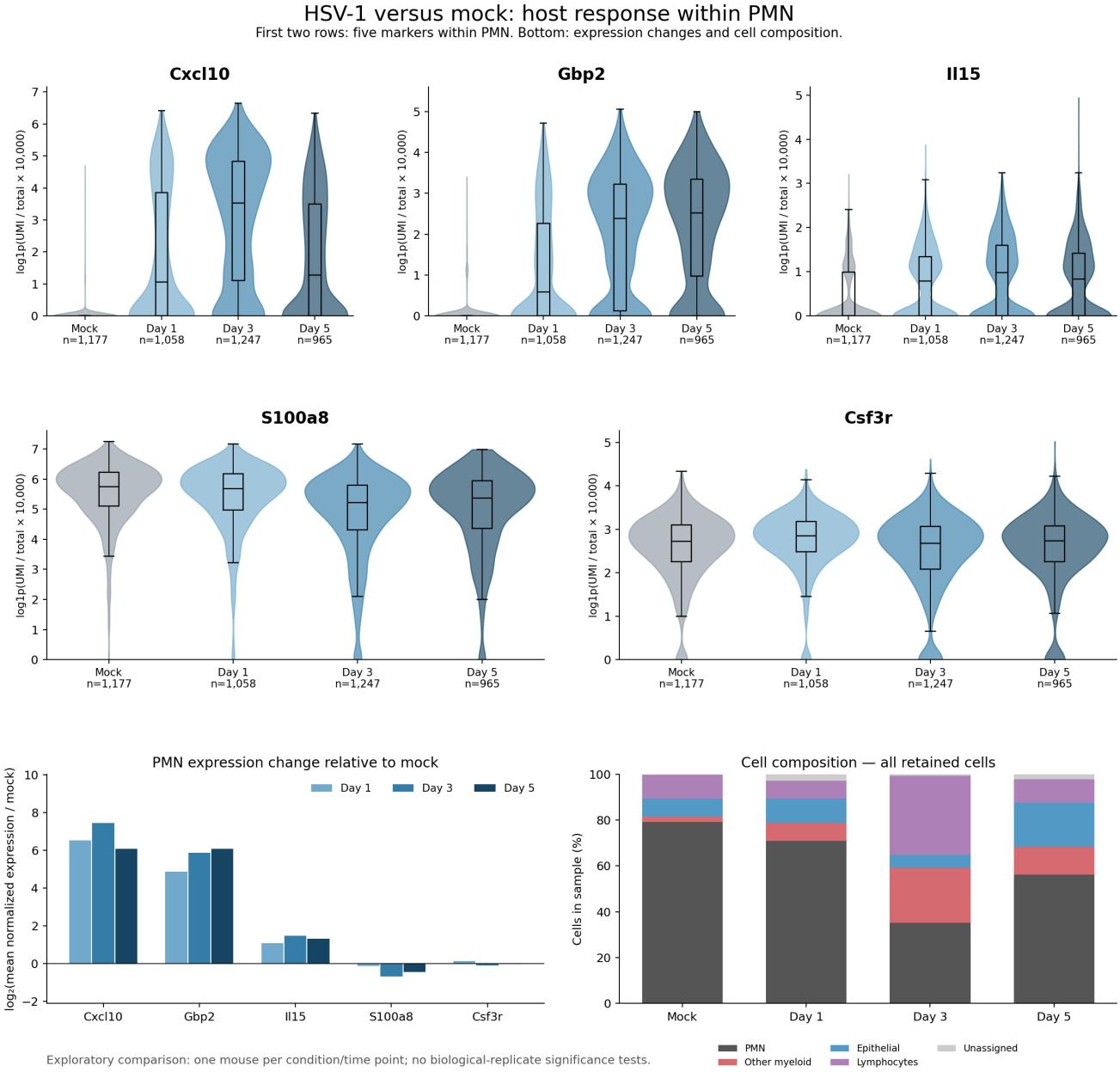
Q4. How do infected and uninfected conditions differ?

I analyzed the mouse UMI counts of the referenced genes per cell. To examine individual gene changes, I plotted **Cxcl10, Gbp2, Il15, S100a8 and Csf3r** within PMN, comparing mock with **days 1, 3 and 5** for each virus. The plots suggest more sustained **Cxcl10/Gbp2** expression in HSV-2 at later time points. This is consistent with the paper, whose **Figure 3D** addresses a related question by showing host IFN-response scores for mock and day-5 infection. Paper, Figure 3.

Marker	HSV-1 pattern	HSV-2 pattern	Interpretation
Cxcl10	Increased by day 1, strongest on day 3, lower on day 5	Low on day 1, high on days 3–5	More sustained late expression in HSV-2
Gbp2	Elevated on days 3–5	Higher expression on days 3–5	Stronger late response in the HSV-2 samples
Il15	Modest changes	Increased on days 3–5	Descriptive trend requiring further validation
S100a8	Expressed in mock and infected-condition PMN	Expressed in mock and infected-condition PMN	Supports PMN identity
Csf3r	Expressed in mock and infected-condition PMN	Expressed in mock and infected-condition PMN	Supports PMN identity

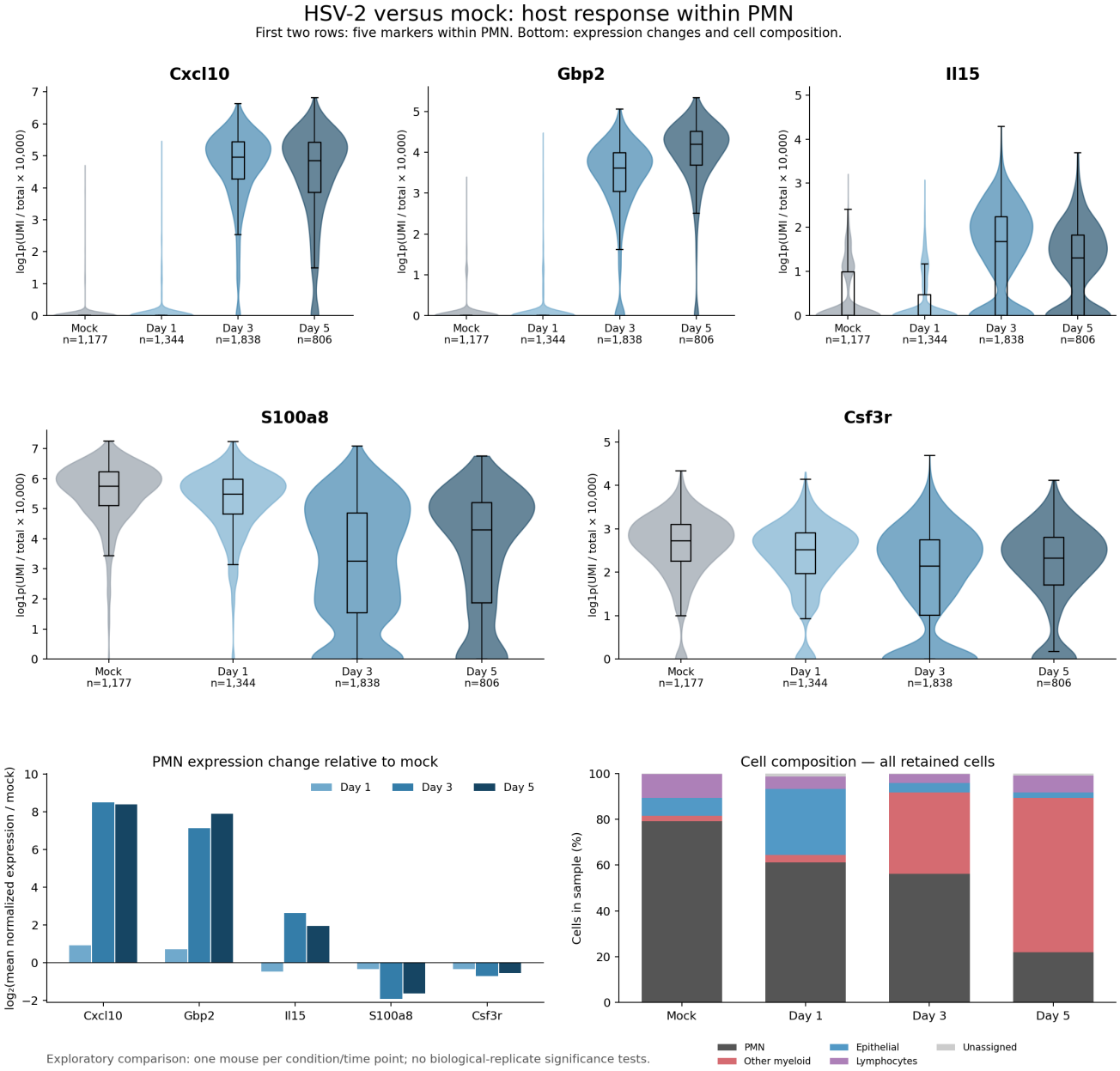
The plots use log-normalized expression within candidate PMN clusters **1, 3, 6, 7, 9, 12 and 13**. These are sample-condition comparisons. With one mouse per condition/time point, the trends do not establish significance across animals.

HSV-1



Exploratory comparison: one mouse per condition/time point; no biological-replicate significance tests.

HSV-2



Exploratory comparison: one mouse per condition/time point; no biological-replicate significance tests.

Gap analysis

Topic	Remaining gap
Sample selection and cell count	The Figure 3 legend describes day-5 tissue. This embedding uses all seven deposited samples, with only mock/day-5 cells displayed in C/D. The paper's 21,633 retained cells are not recovered.
Cluster-based QC	The paper removes three low-gene clusters but does not specify the cluster statistic or initial clustering settings. Our median-gene rule removes four clusters, retaining 17,380 cells.
UMI regression	Negative-binomial regression follows the paper's stated model. In this run, 15 failed fits were handled by Seurat's built-in fallback.
Final clustering	Exact PC count and resolution are unspecified. Our selected settings produce 18 clusters versus the paper's 17. Cluster numbers do not directly correspond to the paper.
Cell-type annotation	The paper names S100a8/Csf3r for PMN but does not list markers for the other Figure 3A groups. Additional assignments remain provisional.
IFN-alpha score	The exact gene-set version, scoring function and control-gene settings are unspecified. Our `AddModuleScore` calculation is an approximation.
Viral counts	HSV-2 alignments are present in inspected BAMs, but its matrix feature is zero in all samples, before and after QC. Mock also has viral alignments and nonzero HSV-1 UMI. The causes remain unresolved.
Biological replication	One mouse per condition/time point limits inference. Cell-level differences do not establish reproducible differences between animals or track individual cell fates.